EIGSEP Memo 003

Marjum Pass 2026-07: pointing

EIGSEP Team

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Abstract

Where the antenna pointed, and how we know. This memo states the mount model and angle conventions, how the campaign pointing table turns motor, potentiometer, IMU and LIDAR readings into azimuth and elevation, what the 07-17 transmitter raster was, and how well its azimuth is known in absolute terms. Its central tests use the ratio of the transmitter's two orthogonal arms, which depends on the antenna's orientation and on nothing else: the antenna beam and the transmitter drive cancel. Every figure, table and quoted number is generated by `make_products.py` in this directory (Appendix A). Positions and the GPS azimuth reference are in memo 002; the observing configuration and timeline are in memo 001.

1 Summary

- **The mount is a roll mount, not alt-az.** Azimuth turns the antenna about its own boresight; elevation then tips the assembly about the highline. Across 16 transmitter tooth pairs (140.6–199.2 MHz) the arm ratio fits the roll mount with a pooled rms of 0.116 (worst pair 0.196) and the best alt-az variant with 0.301 (best pair 0.250; §2).
- **On 07-17 the potentiometer slipped, not the platform.** Measured step by step from the polarization, the platform stays within 3.8° rms (worst 9.0°) of the commanded (motor) azimuth while the potentiometer drifts to −29.2° from it. The arm ratio fits the commanded azimuth better than the potentiometer-fused azimuth on all 16 of 16 pairs (pooled rms 0.116 against 0.162; §4). The table's `az_deg` has been the commanded azimuth since v2.0.
- **The raster** is 20:26:00–21:28:40 UTC on 07-17: after a 180° slew at 20:25:59, the elevation axis turns continuously through full revolutions while azimuth steps 5.000° once per revolution (every 70.8 s), from 190.65° to 90.64°.
- **The table's azimuth zero on the raster is −15.9° from the highline** (±0.7° across pairs), not 0°, from the polarization alone; the HFSS beam fits agree (§5). With the transmitter's arms held at their photographed bearings, only the code's right-handed azimuth fits well, and it needs the arm that memo 001 labels rx6 to have rx1's geometry: the likely error is a swapped arm label, not the handedness.
- **Elevation**: el is the boresight's angle from zenith (0 zenith, +90 horizontal toward bearing 37.84°, 180 nadir). The transmitter is at el = 184.6°. The IMU reads −2.3° from true (§6). Late in the 07-17 raster the box fell up to 33.0° behind the elevation drive on up-legs; the table's elevation, from the IMU, follows the box (§6). The elevation drive failed near 01:41:40 UTC on 07-18, parked at 183.9°, facing the transmitter.
- **Antenna phase center lever arm and flexure**: cross-correlation phase (`box-gnd` × `box-air`) on the transmitter comb through the 07-17 raster sweeps resolves the dipole and balun's 3-D body offset from the elevation axle: $z_{\rm pc} = +55.6 \pm 6.6$ cm along boresight, $x_{\rm pc} = +16.1$ cm along the dipole arm, and $y_{\rm pc} = -1.1$ cm transverse (mean fundamental lever arm 57.5 cm, total $\|\mathbf{r}_{\rm pc}\| = 57.9$ cm; §7). Gravitational flexure adds a 11.3 cm 2nd-harmonic sag when cantilevered horizontally (el = ±90°). Stationary platform wobble is 0.75 cm rms.
- **Table defects** to work around: `EL_POST_FAILURE` is set on every row from 07-14 14:58 onward, including the whole raster (§9).

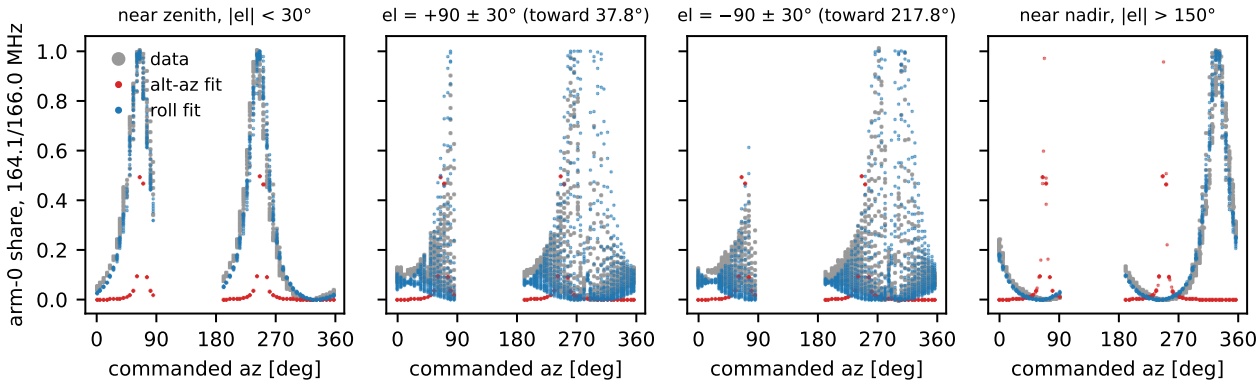


Figure 1: Share of transmitter power in arm 0 against commanded azimuth, in four elevation bands, for one tooth pair (grey), with the best roll-mount (blue) and alt-az (red) fits evaluated at the same samples. The raster covers commanded azimuth 190° through 360° to 91°.

2 The mount and the angle conventions

A vector in the antenna body frame (boresight $+z$, dipole arm $+x$) maps to local ENU as

$$v_{\mathrm{ENU}} = R_z(\psi)\, R_x(\mathrm{el})\, R_z(\mathrm{az})\, v_{\mathrm{body}}, \qquad \psi = 142.164^\circ, \tag{1}$$

implemented as `eigsep_base.rotations.mount_rotation`. ψ is the highline direction counter-clockwise from East (bearing 127.84°/307.84°; memo 002). Azimuth is applied first, about the boresight: **it rolls the antenna and changes only the polarization angle; it never moves the beam.** Elevation is applied second, about the highline, so the boresight always stays in the vertical plane through the antenna at bearing 37.84°/217.84°. The alternative $R_z(\psi)R_z(\mathrm{az})R_x(\mathrm{el})$ (tilt, then turn about the vertical) is the alt-az mount; it is not equivalent under any relabelling.

The antenna and transmitter positions are geometry release `v0004` (memo 002), a point fit with no established uncertainty. Its E, N are UTM grid axes; the line of sight is rotated by the 1.52° grid convergence at the antenna into the true-north frame in which ψ and the transmitter's arm bearings are defined.

Why the beam cannot tell the two apart here. The transmitter is 0.60° from the boresight plane and near nadir, so in either mount its angle from the boresight is close to $|180^\circ - \mathrm{el}|$. Total transmitter power is nearly blind to the mount.

Why the polarization can. The transmitter's two arms are orthogonal horizontal dipoles, driven on alternate comb teeth (memo 001). For a tooth on arm 0 and its neighbour on arm 1, 1.95 MHz above, the fraction of their summed power in arm 0 depends only on the angle between the antenna's dipole and the transmitter's arms, projected perpendicular to the line of sight. The antenna beam is common to both and cancels, as does the absolute drive. Under the roll mount the antenna arm's horizontal projection is $R_z(\psi)\,(\cos\mathrm{az},\ \sin\mathrm{az}\cos\mathrm{el})$: near nadir or zenith it rotates fully with azimuth, toward $\mathrm{el} = \pm 90^\circ$ it is squeezed toward the highline, and the sense of rotation reverses between the upper and lower hemispheres. Under alt-az it is $R_z(\psi + \mathrm{az})\,\hat{x}$ at every elevation, so the arm share would depend on azimuth alone.

Test. For each pair we fit the arm-0 share with three free parameters (an azimuth offset, the bearing of transmitter arm 0, and the arm drive ratio) under each mount and each azimuth handedness, using every good sample of the raster (at least 6442 per pair; the beam product's sample, tooth and slew flags applied). Figure 1 shows one pair; Table 1 summarizes all 16. Per pair, the roll mount fits at 0.084–0.196 rms and alt-az at 0.250–0.319: the best alt-az pair is worse than the worst roll pair. Alt-az cannot make the arm share depend on elevation at fixed azimuth, and the data do. The roll-mount residual is largest at the ends of the band (Fig. 2d), where the antenna's own cross-polarization off boresight, which this model leaves out, is largest.

Table 1: Arm-share fits over 16 tooth pairs (140.6–199.2 MHz). The rms pools every residual of every pair, with the worst pair (for alt-az, the best) in brackets; angles are means weighted by sample count, with the range over pairs. "Az zero from highline" is the offset added to the table's azimuth so that az = 0 puts the dipole along the highline; "arm-0 bearing" is the compass bearing of transmitter arm 0, mod 180°. The last row repeats the preferred model on the potentiometer-fused azimuth (§4).

mount	handedness	pooled rms (worst pair)	az zero from highline	arm-0 bearing
roll	right-hand	0.116 (0.196)	−15.9° (-16.9 to -14.7)	79.7° (76.5–83.4)
roll	left-hand	0.118 (0.196)	+16.0° (+14.6 to +17.1)	175.8° (172.2–178.9)
alt-az	right-hand	0.301 (0.251)	—	—
alt-az	left-hand	0.301 (0.250)	—	—
roll, pot-fused az	right-hand	0.162 (0.230)	+4.0° (+2.3 to +4.6)	69.6° (66.1–72.6)

Elevation convention. el is the boresight's angle from zenith by the right-hand rule about the highline pointing west-north-west (bearing 307.84°): 0 is zenith, +90 is horizontal toward 37.84°, 180 (equivalently −180) is nadir. The table reports el in $(-180, 180]$, so the elevation axis turning through nadir shows as a jump from +180 to −180. Beware the wording "180° from zenith" for the transmitter: it sits at el = 184.6°, which the table writes as 184.6° − 360°, not at nadir.

Azimuth convention. az is by the right-hand rule about the boresight (about up when el = 0), so increasing az turns the dipole counter-clockwise seen from above at el = 0, and clockwise seen from above when the antenna faces down. At az = 0 the dipole lies along the highline *by definition*. Whether the table's az = 0 meets that definition is the question of §5. The handedness is the geometer's 2026-09-18 reading of the hardware. The transmitter polarization alone cannot test it, but with the transmitter's photographed arm bearings it favours this handedness (§5).

3 The sensors and how the table combines them

The pointing table (`marjum-2026-07/curation/pointing_table.parquet`, v3, 1,227,956 rows from 5,120 files) has one row per correlator sample, joined on `t_utc`. Its README documents the columns, the clock validation, and coverage by era; this section records the rules and what is known about each sensor. The streams are in each correlator file's `metadata` group, natively aligned with the spectra.

Motor counts (az and el, 180/11300 deg per step). Smooth and finely quantised, but relative: they re-home with jumps of up to ∼46°, and they say where the drive was told to go, not where the platform went.

Potentiometer (az). The only absolute azimuth sensor. Scatter 1.73° on a static platform, and it slips: through the 07-17 raster it drifted from −2.6° to −29.2° against the platform (§4).

IMU elevation. Gravity-referenced and absolute, scatter 0.24°; read −2.3° from true (§6).

IMU azimuth. Dead from 07-16 on, and drifting at ∼190°/hr where alive. Not used.

LIDAR. A rangefinder mounted 90° from the boresight; it points at nadir when el = +90. 46% of its 52,742 readings are the 250 m out-of-range sentinel. It is a geometry check, not an altimeter (§6, §8).

Elevation is the IMU wherever it is finite. Where it is not, the motor fills in, flagged `EL_NO_IMU` and `suspect`. Where the elevation moves faster than 20°/s between adjacent samples, both samples are flagged `EL_SOLUTION_GLITCH` and `suspect`, since the rate does not say which one is wrong. On the raster 146 rows are flagged, near el = ±180°, among them 60 onboard-solver jumps of ±120° or 180° in `imu_el_deg`; every such jump is flagged.

Azimuth (`az_deg`, v2.0) is the *commanded* azimuth: motor counts give the motion, and the potentiometer sets one absolute offset per anchor segment. A new segment starts at every re-home (a count jump > 10° between samples), every telemetry gap > 5 s, and every big slew (continuous motion spanning > 20°; the raster's 5° steps never qualify). The offset is the median of pot − motor over the first 10 minutes of the segment after its

slew. There are 165 segments campaign-wide. The rule rests on two facts: the az and el drives never run at the same time and are locked when idle (Aaron), and on 07-17 the platform demonstrably followed the motor (§4). The v1.x azimuth (a complementary filter re-anchoring the motor to a ±30 s rolling median of the potentiometer) survives as `az_fused_deg`. Code: `data-analysis/notebooks/arp/marjum-2026-07/pointing/fuse.py` (`commanded_azimuth`, `fuse_azimuth`, `fuse_elevation`).

What commanded azimuth assumes. That the platform follows the motor between big slews. That is tested only on the 07-17 raster. Absolute azimuth comes from one potentiometer reading per segment, so it is no better than the potentiometer's zero at that moment, and segments either side of a slew can differ by whatever the platform slipped during it.

Height is the era's nominal value for almost every row, replaced by the LIDAR range when that is 20–200 m and the antenna faces down. Field heights are in three different datums (memo 002, Heights); this table does not reconcile them.

Table 2: Flag bits in pointing table v3: rows set campaign-wide, and the share of the raster (20:26:00–21:28:40) on which each is set. `AZ_SLIP_EVENT` and `AZ_SLIP_RAMP` now mark potentiometer/motor divergence, not platform slip.

bit	name	rows, campaign	share of raster
1	`NO_METADATA`	442,080	0%
2	`AZ_NO_POT`	458,800	0%
4	`AZ_SLIP_EVENT`	17,701	1%
8	`EL_NO_IMU`	485,396	0%
16	`EL_STUCK`	4,410	0%
32	`UNCOMMANDED_MOTION`	9,560	0%
64	`NO_ESTIMATE`	464,359	0%
128	`HEIGHT_ASSUMED`	1,227,447	100%
256	`EL_POST_FAILURE`	700,556	100%
512	`AZ_SLIP_RAMP`	27,876	20%
1024	`EL_SOLUTION_GLITCH`	983	2%
2048	`AZ_POT_DIVERGES`	26,499	71%
4096	`AZ_SLEW`	3,677	1%
8192	`AZ_FUSED_FALLBACK`	8,911	0%

4 The 07-17 transmitter raster

The motor's azimuth record fixes the raster's structure exactly (Fig. 2). At 20:25:59 the azimuth slews by 180° to 190.65°. From 20:28:12 to 21:28:23 it then steps 52 times, every step 5.000° on average (4.986–5.002, the motor quantisation), one per elevation revolution, at a mean interval of 70.8 s, ending at 90.64°: 260.0° in all. Between steps the elevation axis turns continuously through a full revolution, so each revolution passes the transmitter once near el = 184.6°. The beam fits use 20:26:00–21:28:40. Read from the potentiometer, the same motion looks like uneven steps of ∼6° ending near 55°, which is how it was first described.

The potentiometer slipped, not the platform. The potentiometer does not track the motor (Fig. 2b): their difference goes from −2.6° at the start to −29.2° at the end, falling fastest (−154°/hr) over 20:41–20:54, which v1 of the table recorded as a −27.35° platform slip. Either the platform lagged the motor or the potentiometer lagged the platform. The polarization decides: the arm share follows the commanded azimuth (pooled rms 0.116) better than the potentiometer-fused azimuth (0.162) on every pair (Fig. 2d).

The direct test is to measure the platform's azimuth from the polarization one step at a time. Holding each pair's fitted arm bearing, drive ratio and offset fixed, we fit one extra azimuth shift per raster step to all pairs together (Fig. 2b, blue). Over 52 steps the shift has rms 3.8° and never exceeds 9.0°; inside 20:41–20:54, while the potentiometer falls to −28.7°, the median |shift| over its 11 steps is 1.5°, no larger than the 4.0° median over the other steps. The largest single-step shift in that window is not quoted: it changes by several degrees under

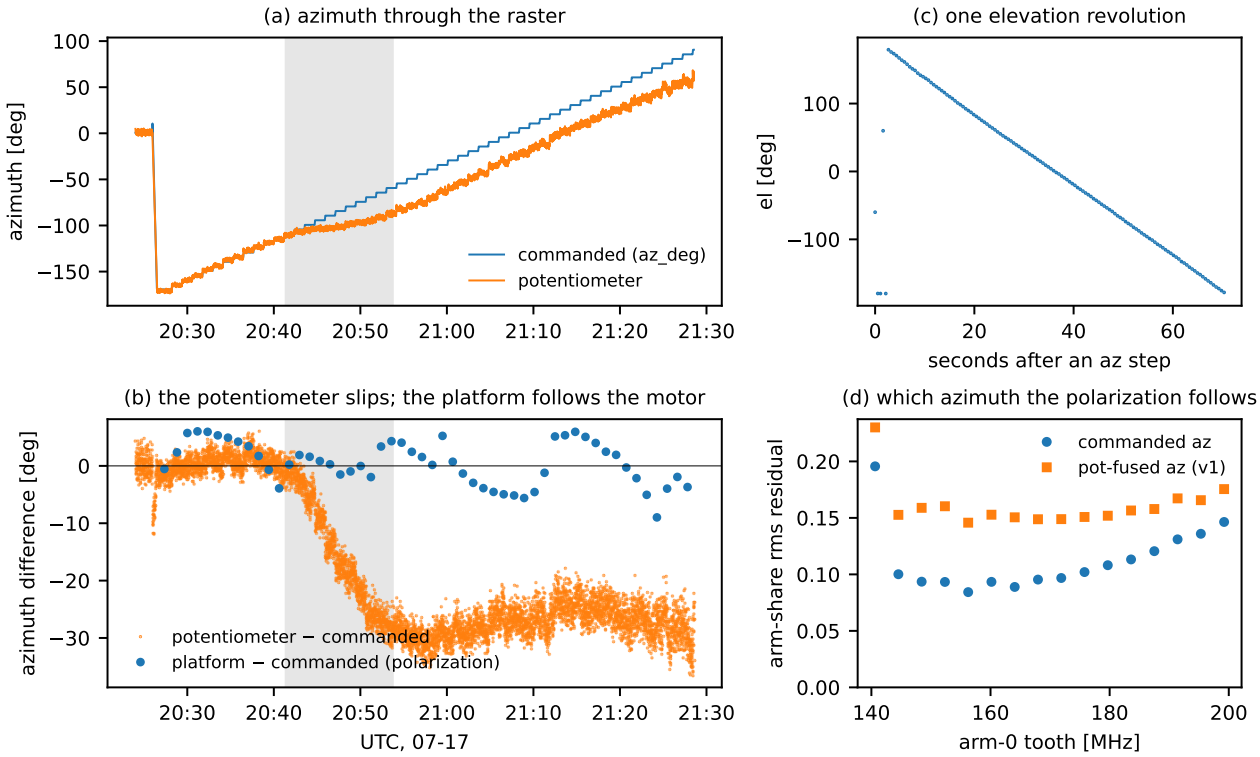


Figure 2: (a) Commanded and potentiometer azimuth through the raster. (b) Their difference; the shading is the interval v1 of the table treated as a $-27°$ platform slip. (c) Elevation through one revolution between azimuth steps. (d) The arm-share residual of the roll-mount fit (§2) per tooth pair, using commanded azimuth and using the potentiometer-fused v1 azimuth.

sub-degree changes in the assumed antenna and transmitter positions. **The platform followed the motor.** The smooth few-degree wander of the shift is a bound on platform-motor differences and model error together.

By time, pooled over pairs:

pooled arm-share rms	before 20:41:24	20:41:24–20:53:43	after
commanded azimuth	0.073	0.175	0.103
potentiometer-fused azimuth	0.138	0.203	0.155

Both azimuths fit worse inside 20:41–20:54 and the minutes after. Since the per-step shift stays small there, the extra scatter is not pointing; its cause is not identified. This agrees with the beam-model test that led to v2.0 (data-analysis `e0ffa78`, `notebooks/arp/marjum-2026-07/single_ch_beam_fit.ipynb`): the unfitted HFSS beam's held-out test residual at 156.25 MHz is 0.071 with commanded azimuth against 0.197 with v1. The present test does not use the beam model at all.

5 Absolute azimuth

Memo 002 places az = 0 (dipole along the highline) at bearing 37.84° for the boresight plane, from the GPS anchors, ±0.6°; that is the geometry of the highline and is not in question. What is in question is whether the table's `az_deg` = 0 puts the dipole along the highline. It inherits the potentiometer's zero, one reading per segment, and the potentiometer slips.

The polarization measures it. The roll-mount fit's azimuth offset is not degenerate with the transmitter's arm bearing, because the cos el factor in the arm's projection is referenced to the highline, which is fixed. In the right-handed convention of Eq. (1) it gives $-15.9°$ ($\pm 0.7°$; -16.9 to -14.7 across pairs) on the raster's

segment, and puts transmitter arm 0 at bearing 79.7° (76.5–83.4). The HFSS beam fits, in the same convention (`derived/beam/empirical_raster_v0010` onward; the reference beam is now `v0013`), independently want −16.0° (`table_az_zero_from_highline_deg`) and $\alpha \approx 96°$, an arm-0 field along bearing 84° (memo 001).

Free fits cannot choose the handedness. Reversing the azimuth handedness (az → −az in Eq. (1)) fits as well (0.118 against 0.116; Table 1), with the offset of the opposite sign (+16.0°) and arm 0 mirrored about the highline, to 175.8°. Each solution is the mirror image of the other.

The photographs can. Memo 002 measures the transmitter's rx6 axis at 165.5° true from the feed's corners in IMG_2203 (sd 0.74° over the sampled v0004 poses; `derived/tx_feed_orientation/v0002`), and rx1 at 75.5°. With arm 0's bearing held at one of these, the polarization fixes the table azimuth at which the antenna's dipole lines up with each arm, and the handedness then fixes which way the dipole turns from az = 0 to get there: the two handednesses imply different zeros, and the off-nadir elevations test each one independently of the bearing. Table 3 gives all four combinations.

Table 3: Arm-share fits with transmitter arm 0's bearing held at the photographed rx6 or rx1 axis (memo 002), for each azimuth handedness; only the azimuth offset and the drive ratio are free. The free fits of Table 1 are repeated for reference.

arm 0 is	handedness	az zero from highline	pooled rms (worst pair)
rx6 (165.5°)	right-hand	+75.3° (+72.4 to +77.2)	0.218 (0.259)
rx6 (165.5°)	left-hand	+19.9° (+17.7 to +24.8)	0.173 (0.245)
rx1 (75.5°)	right-hand	−14.7° (-16.7 to -11.9)	0.129 (0.201)
rx1 (75.5°)	left-hand	−69.9° (-72.4 to -64.6)	0.247 (0.289)
free	right-hand	−15.9°	0.116
free	left-hand	+16.0°	0.118

Only one combination fits nearly as well as the free fit: **right-handed azimuth, with arm 0 along rx1** (rms 0.129, against 0.116 free; better than the next-best combination on 15 of 16 pairs), giving a zero of −14.7°. Arm 0 along rx6 requires the left-handed azimuth, with a zero of +19.9° and a clearly worse fit (0.173): the data put arm 0 10.2° from the photographed rx6 axis, and the zero cannot move to absorb it. The other two combinations imply zeros of +75.3° and −69.9° and fit worst.

So the arm labels are the likely error. Three statements cannot all hold: the photographed arm geometry, the right-handed azimuth of `eigsep_base`, and memo 001's identification of arm 0 (teeth on channels ≡ 0 mod 16) as rx6. The photographs and the right-handed azimuth agree with each other and with the data to 4.2°. The arm identification rests on a chain of labels: the field display's "rx6" during 15:54:53–15:56:22 on 07-17, the cable from that output to the feed, the quadrant pair marked "6" in IMG_2204, and that pair's axis in IMG_2203. One swap anywhere in the chain reconciles everything, while a wrong handedness leaves an unexplained 10.2°. This memo therefore adopts the right-handed azimuth and a table zero of −15.9° on the raster, and takes the arm that radiated alone at 15:54:53 to lie along the axis photographed as rx1. The beam fits' $\alpha \approx 96°$ (bearing 84°) is the same result. Either way, **the table's zero on the raster is ∼15–20° from the highline, not 0**.

Other epochs. The 16° applies to the raster's anchor segment only. The LIDAR terrain fit (memo 002, Azimuth zero) uses the 07-18 01:37 sweep, a different segment and a different potentiometer state, and finds the zero within 1.1° of the highline; nothing requires the two to agree, given a potentiometer that slipped ∼27° in an hour. For other segments, no absolute azimuth measurement exists, and their `az_deg` should be read as "potentiometer zero, to within tens of degrees".

6 Elevation: zero, the LIDAR, under-travel, and the failure

Zero. The LIDAR is mounted 90° from the boresight and points at nadir when el = +90. Its 492 returns at 85–100 m within 10° of el = +90 (the long dwell at the final height) have mean range 92.12 m and mean IMU elevation 87.7° (Fig. 3a). If the LIDAR is exactly 90° from the boresight, the IMU reads −2.3° from true (about

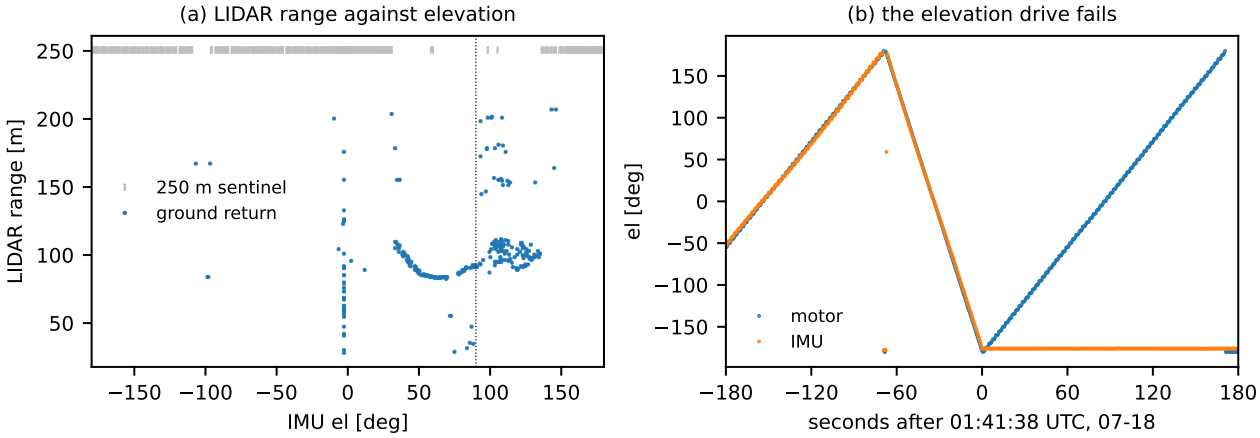


Figure 3: (a) LIDAR range against IMU elevation for every LIDAR reading in the campaign. Ground returns span el 0–150°, with the long dwell at el ≈ 88°, where the LIDAR points at nadir; facing up it reads the 250 m sentinel. (b) Motor and IMU elevation around the 07-18 failure: the motor keeps sweeping while the IMU stays fixed.

−2.3° with the boresight at zenith). That angle is not measured, so this route gives the IMU zero plus the LIDAR's mounting error. The reference beam fit (`v0013`, on geometry release `v0004`) wants only +0.93° near the transmitter, with the transmitter offset pinned at the fit's ±1.06 m bound in east and north, which can absorb about 0.9°. That bound was set from the historical `v0001` bracket; v0004 has no established uncertainty. The two routes are not yet reconciled (§10). The *sign* and origin of el are not in doubt: the LIDAR sees ground with el between 0 and 180 and almost never facing up (el near −90), the right-hand rule about the west-pointing highline gives the same, and the corrected LIDAR geometry puts nadir at el = +90 (legacy MEMO-012 §4.1, in git history before this memo).

Under-travel late in the raster. From leg 45 on, the box turns less than the elevation drive's step count on up-legs (Fig. 4): the counts advance at the normal rate, but the box lags them. It does not stall, except momentarily on leg 45. The counts are open-loop (§3), so they cannot tell steps the motor missed from a slip between the motor and the box. The drive angle is `motor_el_deg`, unwrapped in time; the raster's 54 legs run between its reversals. The residual is `imu_el_deg` minus the drive angle, with the constant offset between their zeros subtracted (−2.03°, the median over legs 0–44), and the 60 onboard-solver jumps removed (§3). Changes are taken in each leg's direction of travel, so a box that turns less than the drive gives a negative change on an up-leg and a positive one on a down-leg. Through leg 44 the residual's leg medians stay within ±4.1°. Afterwards the loss is concentrated at drive angles 90–150°: across that window up-legs 45, 47, 49 and 51 lose 13.3, 4.5, 5.6 and 5.7° more than the median of the 22 earlier up-legs, or 12.1, 4.1, 5.1 and 5.1 times their standard deviation. Over the rest of the sweep their excess is −2.4 to +1.6°, within 1.3 standard deviations; leg 45 loses part of it before the window and regains part after (Fig. 4b). Down-legs show no comparable loss (Fig. 4c): across the same window the late down-legs change by +0.2 to +1.7° more than the earlier ones, at most 1.2 standard deviations. Nor do they give the up-legs' loss back: their net change over a sweep, +1.3 to +3.9°, is within 1.3 standard deviations of the earlier down-legs' +1.5 ± 1.9°, where a recovery would raise it by the up-legs' loss. Leg 45 nearly stalls 3 times: in single 0.54 s integrations the drive advances 2.9° while the box moves −0.3 to +0.6°, which is 58% of its window loss. On the later legs the box keeps moving. No integration loses more than 1.4°, and over the 34–36° of drive angle in which 10–90% of the loss accrues, the box turns 10–13% less than the drive, relative to the earlier legs. The window was found on these legs, so a trials factor applies: scanning 60° windows along each earlier up-leg, 2 of the 22 reach leg 47's z somewhere, and none reaches those of the other three. By leg 52, the last full leg, the box lags the drive by 28.5° (at most 33.0°, on leg 51), and 1,295 of 6,930 raster rows (18.7%) are more than 5° off.

The table's elevation is the IMU's, so these rows are pointed correctly; `motor_el_deg` is not a pointing on them, and the raster is not a uniform sweep there. The cause is not established. A mechanical bind, a loss of drive torque, an azimuth-dependent load and supply-voltage sag remain unresolved possibilities.

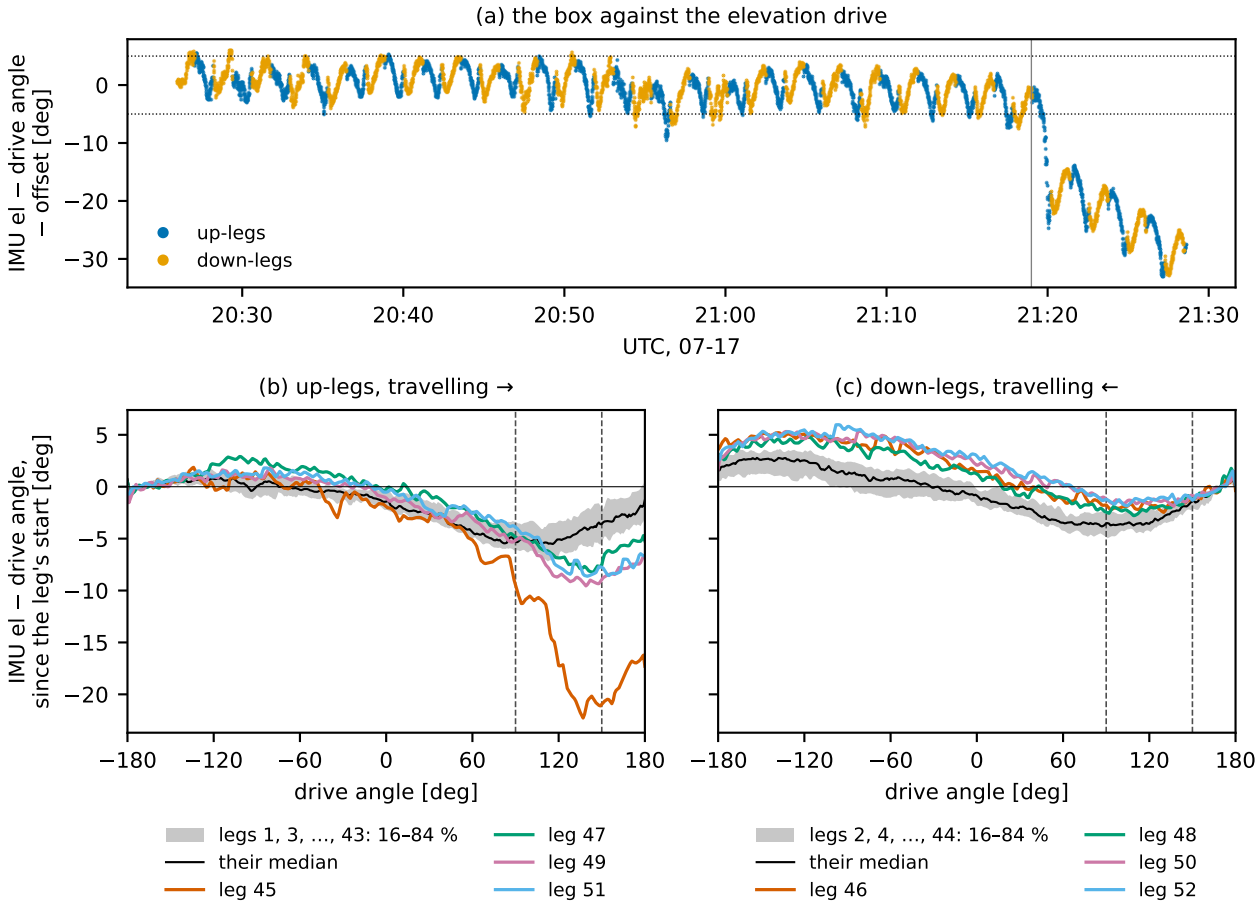


Figure 4: The box against the elevation drive through the 07-17 raster (20:26:00–21:28:40 UTC, pointing table v3; the 60 solver-jump rows removed). (a) The box's elevation from the IMU (`imu_el_deg`) minus the drive angle from the step counts (`motor_el_deg`, unwrapped), on every row, coloured by the leg's direction of travel. The constant offset between the two zeros, −2.03° (the median over legs 0–44, where the box follows the drive), is subtracted, so 0 means the box is where the counts put it and negative values mean it is behind. Dotted lines at ±5°; the grey line marks the start of leg 45. From there the box falls behind on up-legs and keeps the lag on down-legs. (b) Along each up-leg, which travels left to right: the same quantity against drive angle, relative to its level where the leg starts (its median over the first 20°), so a falling curve means the box is falling further behind the counts. Grey: the 22 earlier up-legs (16–84 % range, and median). Even healthy legs are not flat, so the late legs are compared with them, not with zero. Colours: the late up-legs. Dashed lines: the 90–150° window over which the text's numbers are computed. The late up-legs drop away from the band between them, and how far they drop below the band's own change across the window is the excess quoted in the text. (c) The same for down-legs, which travel right to left from +180°: the late down-legs keep the earlier legs' shape, with no comparable loss in the window (§6).

What the current monitor shows. The shared supply current detects the elevation drive: on every late leg it draws 0.24–0.29 A above the nearby parked background, so the drive remains energised during the under-travel. The pooled running-minus-parked estimates are 0.273 A on early legs and 0.264 A on late legs. With both axes parked, the background stays near 2.493 A: its fitted change over the raster is $+7.2 \pm 3.9$ mA, and the 5 parked samples in the final 15 minutes depart from the earlier trend by only $+2.7 \pm 4.2$ mA. These are no detected changes; the sparse parked samples do not bound brief excursions. After background subtraction, the downward sweeps' draw has a gradual trend of -36 ± 4 mA/h, but neither direction shows a resolved additional step when the under-travel starts (combined $+4.9 \pm 4.2$ mA). All quoted errors here are one standard error; the method is in Appendix A.

For orientation-dependent load the comparison is at matched IMU elevation. In the 90–150° IMU-angle window, the additional late-leg current contrast is $+15 \pm 25$ mA: no increase is detected. A window rise already exists on healthy up-legs (71 ± 10 mA in drive angle). Using drive angle instead gives an additional late rise of $+68 \pm 25$ mA, weak evidence; a direct paired test does not establish that the two coordinate comparisons differ. Matching elevation alone does not control azimuth or time. The *load thermistor* rises from 33.14 to 34.92°C on up-legs 1–51, while time and azimuth advance together, so this raster cannot separate the three as onset markers. It is not a measurement of motor or driver temperature. No battery voltage was recorded: approximately steady current alone cannot exclude supply-voltage sag. Whether the under-travel was an early stage of the later freeze remains open.

The failure. The last successful downward sweep completes at 01:41:39.38 UTC on 07-18. The next leg's step counts begin advancing at 01:41:40.45, 1.07 s later, with no comparable drive-current draw and the IMU elevation fixed (Fig. 3b). Across the 6 frozen legs in the two selected post-re-raise blocks, the mean extra draw is 1.1 ± 0.5 mA, or 0.34% of that morning's 0.313 A continuous running draw (0.7% at mean plus two standard errors). Thus there is no measurable elevation-drive current at the normal running scale, although the counts advance. This differs from the energised drive during the under-travel; it does not identify the failed component or establish a causal link between the events.

The antenna stayed near −176.05° (183.9°), facing the transmitter, to the end of the data at 03:08:49. The stillness criterion used for this memo's Fig. 3b first fires at 01:41:38 UTC; it is a detector boundary, not the physical failure instant. The current-and-command comparison above brackets the loss of response between sweeps. The field notes give 01:55, and legacy MEMO-002's trailing detector gave 01:42:32.

7 Antenna phase center lever arm, platform stability, and elevation flexure

The mount model (Eq. 1) maps vectors from an idealized antenna body frame to local ENU. In the physical assembly, the antenna electromagnetic phase center (the dipole feed point and balun) does not sit on the mechanical gimbal axle: it is displaced by an offset vector $\mathbf{r}_{\mathrm{pc}} = [x_{\mathrm{pc}}, y_{\mathrm{pc}}, z_{\mathrm{pc}}]^T$ in the antenna body frame (boresight $+z$, dipole arm $+x$). Because the transmitter sits near nadir at el $\approx 184.6°$, the unit line of sight from the antenna to the transmitter is nearly vertical, $\hat{\mathbf{k}}_{\mathrm{tx}} \approx [0.046, 0.045, 0.998]$ in ENU. Under the roll mount rotation $R(\mathrm{az}, \mathrm{el}, \psi) = R_z(\psi) R_x(\mathrm{el}) R_z(\mathrm{az})$, the line-of-sight path length to the transmitter modulates as

$$\Delta d(\mathrm{el}, \mathrm{az}) = \hat{\mathbf{k}}_{\mathrm{tx}} \cdot R(\mathrm{az}, \mathrm{el}, \psi)\, \mathbf{r}_{\mathrm{pc}} \approx z_{\mathrm{pc}} \cos(\mathrm{el}) + [x_{\mathrm{pc}} \sin(\mathrm{az}) + y_{\mathrm{pc}} \cos(\mathrm{az})] \sin(\mathrm{el}) + d_0. \tag{2}$$

A complete elevation revolution sweeps cos(el) between +1 (zenith) and −1 (nadir), producing a peak-to-peak path modulation of $2z_{\mathrm{pc}}$. The transverse terms $x_{\mathrm{pc}}, y_{\mathrm{pc}}$ modulate this path with azimuth as the antenna rolls.

Multi-frequency geometric consistency. The cross-correlation between the stationary ground antenna (`box-gnd`) and the raised antenna (`box-air`) on the transmitter comb teeth (memo 001) yields an interferometric phase $\phi(\nu) = 2\pi\nu\, \Delta d/c + \phi_0$. Across 187.5–218.75 MHz (channels 768 to 896), the phase excursion per elevation sweep scales strictly with frequency: on reference file `corr_20260717_203032Z.h5`, the unwrapped phase span increases from 313.5° at 187.5 MHz to 377.7° at 218.75 MHz, yielding an invariant geometric path modulation of 139.3–143.8 cm peak-to-peak (Fig. 5a). This confirms that the modulation represents non-dispersive path geometry, not an electronic dispersion or instrumental phase artifact.

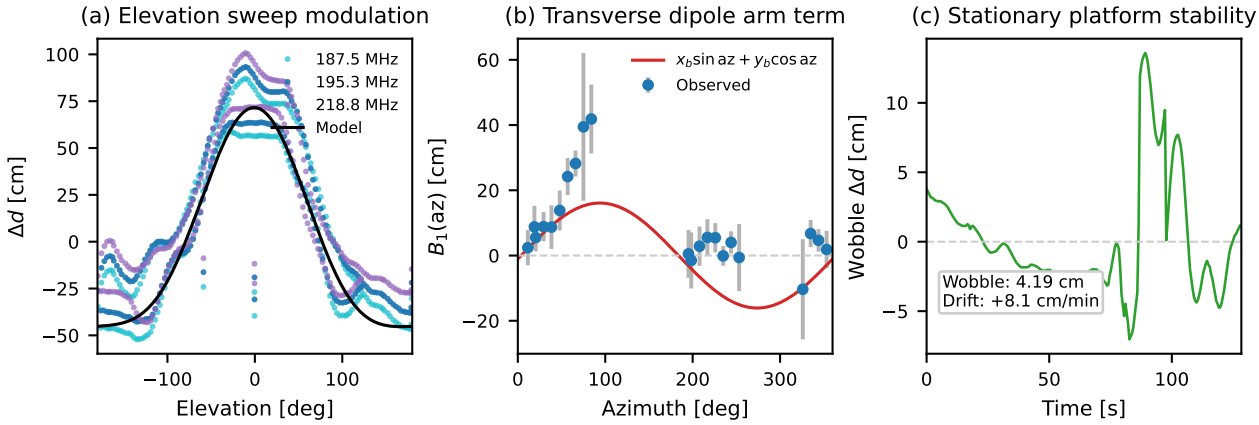


Figure 5: (a) Path length modulation Δd along the line of sight to the canyon transmitter across three comb frequencies (187.5 MHz, 195.3 MHz, and 218.8 MHz) during one elevation sweep of the 07-17 raster (`corr_20260717_203032Z.h5`, az $\approx 198°$), with the fitted 3-D mount model (black). (b) Fitted transverse lever arm term $B_1(\mathrm{az}) = x_b \sin \mathrm{az} + y_b \cos \mathrm{az}$ against commanded azimuth across the 22 clean raster sweeps (error bars: residual path rms), confirming the +16.1 cm offset along the dipole arm ($+x_b$) and near-zero (−1.1 cm) transverse offset ($+y_b$). (c) Platform stability during a 128 s stationary dwell (`corr_20260717_155704Z.h5`): linear thermal/cable drift (+8.1 cm/min) subtracted, leaving high-frequency wind wobble of 0.75 cm rms on < 5 s timescales (4.19 cm over the 128 s dwell, 9.8° phase rms at 195.3 MHz).

3-D lever arm measurement. Fitting Eq. (2) across the 22 clean raster sweeps outside the cross-polarization null (Table 1) decomposes the 3-D phase center position:

- **Boresight offset** z_{pc}: $+55.6 \pm 6.6$ cm along $+z$. When pointing at zenith, the dipole sits ≈55.9 cm above the elevation axle, furthest from the canyon floor; at nadir, it swings ≈55.9 cm below the axle, closest to the transmitter.
- **Dipole arm offset** x_{pc}: +16.1 cm along $+x$ (along the dipole axis).
- **Transverse offset** y_{pc}: −1.1 cm along $+y$, consistent with zero within measurement scatter. This matches the mirror symmetry of the dipole and balun mounting.

The total 3-D lever arm is $\|\mathbf{r}_{\mathrm{pc}}\| = 57.9$ cm, with a mean fundamental lever arm of 57.5 ± 8.7 cm (median 55.9 cm) across sweeps. This directly accounts for the physical placement of the dipole elements, balun, and feed hub on the mast, which sit 40–55 cm from the elevation axle.

Gravitational flexure vs. elevation. Decomposing the elevation path residual into higher Fourier harmonics reveals a persistent 2nd harmonic of amplitude 11.3 ± 2.9 cm, dominated by the cosine term $A_2 \cos(2\mathrm{el}) = +9.6$ cm. At zenith (el = 0°) and nadir (el = 180°), gravity acts along the boresight mast, producing minimal lateral bending. At horizontal pointings (el = ±90°), gravity acts perpendicular to the mast and balun harness, deflecting the cantilevered structure downward toward nadir by ∼11.3 cm relative to the rigid circle.

Platform stability. During continuous motor-driven sweeps, the residual path rms after removing the 3-D lever arm and 2nd-harmonic flexure is 7.0 cm. In stationary dwells (Fig. 5c, `corr_20260717_155704Z.h5`), the antenna experiences high-frequency wind wobble of only 0.75 cm rms on < 5 s timescales (4.19 cm rms over the full 128 s integration, corresponding to 9.8° of phase at 195.3 MHz), alongside a steady thermal/cable drift of +8.1 cm/min.

Horizon model modulation vs. elevation. The 55.9 cm phase-center offset from the elevation axle shifts the observer's position through which the terrain skyline is seen. For a skyline ridge at horizontal range $r(\phi)$ and elevation angle $\theta_0(\phi)$, the first-order horizon elevation change is $\Delta\theta(\phi; \mathrm{el}) \approx -[\Delta U(\mathrm{el})/r(\phi)] \cos^2 \theta_0(\phi)$. On the current DEM (`derived/dem/v0004`), the skyline ridges stand at ranges from 158.8 m to 7.1 km (median 370.2 m). As the antenna rotates from nadir (calibrations) to zenith (science observations), the phase center rises by $2z_{\mathrm{pc}} \approx$

1.11 m, lowering the apparent horizon across the entire sky by a median of 0.112° (6.7 arcmin) and up to 0.389° (23.3 arcmin) on the closest ridge. Relative to the nominal axle position (`curation/horizon_profiles_v0004`), zenith pointing lowers the horizon by −0.067° (median; worst −0.205°), while nadir pointing raises it by +0.046° (worst +0.274°). At a typical solar elevation rate of ≈ 0.15°/min, this modulation shifts the predicted geometric sunrise and sunset ridge crossings by up to 1.4 min. Integrating the Global Sky Model (GSM) through the HFSS beam over DEM `v0004` terrain, this modulation changes the predicted antenna temperature by +2.6 K (+5.3 K peak-to-peak) at 50 MHz, falling to 30–50 mK at 150 MHz and −0.1 K at 200 MHz where the ground temperature ($T_{\rm gnd} = 290$ K) exceeds the sky temperature.

8 The LIDAR-code fixes

The LIDAR terrain constraint predicted LIDAR ranges from the DEM with three defects, all fixed on 2026-09-18 in the since-retired `terrain` repository. The corrected constraint is the published product `marjum-2026-07/derived/lidar_constraint/v0001`, whose manifest records its code and inputs, and is what memo 002 reports; it was computed at the historical `v0001` antenna position.

1. **Sign** (`319f3e9`): the ray bearing is $\Delta_{\rm az} - {\rm az}$, not ${\rm az} + \Delta_{\rm az}$, because bearing decreases as az increases.
2. **Mount** (`4b63b01`): the rays were computed with the alt-az mount. The two mounts agree only at ${\rm az} = 0$, which is why the old code still reproduced the dwell plateau. The fix uses Eq. (1) and also removes a double shift of the DEM origin.
3. **Retracted constant**: a nadir elevation of 92.25°, already retracted, was still live. The corrected mount puts nadir at ${\rm el} = +90$ by construction and replaces both competing nadir constants by a fitted elevation offset δ (also `4b63b01`).

Later commits fix $\Delta_{\rm az}$ at the GPS value 37.836° instead of fitting it (`e153ba4`), cite the free-$\Delta_{\rm az}$ scan instead of rerunning it (`371670e`), check that scan's provenance (`0acb255`), and stop a double-anchoring of the DEM cache (`6205128`).

9 Known defects in pointing table v3

- **`EL_POST_FAILURE` never clears.** It latches at the first `EL_STUCK` detection, on 07-14 14:58 (the elevation drive's set screw was replaced on 07-16; memo 001), and stays set on every later row: 57% of the table and all of the raster (Table 2). Do not use it to select scanning data until it is rebuilt to reset after a repair; use `EL_STUCK` and the time of §6.
- **`az_motor_pot_offset_deg`** holds pot−motor, the opposite of its name.

10 Open questions

- The transmitter's arm labels. Tracing which feed quadrant pair the display's "rx6" output drives (field notes, IMG_2203/2204, the cabling) would confirm or rule out the swap that §5 infers; a bench check of the potentiometer's sense would test the handedness directly.
- Absolute azimuth outside the raster. Any epoch with the transmitter on and the antenna rolling would give it by the method of §5; the 07-16 transmitter-on era is a stare and cannot.
- Whether the platform follows the motor in the earlier eras, where the potentiometer and the commanded azimuth disagree most (95th percentile ∼30° in the ∼2 m and ∼91 m eras).
- The IMU elevation zero: the LIDAR route gives −2.3°, the beam fit +0.93°. A measurement of the LIDAR's angle to the boresight, or a fit with the transmitter position held at the v0004 value, would separate them.
- The tether anchor's position, needed to put the 87.5 m and 91 m heights on a datum (memo 002).
- The cause of the late-raster under-travel (§6), and whether it led to the 07-18 failure.

Revision history

2026-10-01 First version. Replaces legacy MEMO-002 (pointing table v0) and MEMO-013 (fusion logic), and the mount, elevation and LIDAR-code parts of MEMO-012. New in this memo: the polarization tests of the mount, of commanded azimuth, and of the azimuth zero; the exact raster structure from the motor record; the elevation failure time from the data. Same day, after review: fits with the transmitter's arms held at their photographed bearings favour the right-handed azimuth (zero $\approx -15°$) and a swapped arm label over a left-handed azimuth, reversing the first draft's lean.

2026-10-15 Added the late-raster elevation under-travel (§6): from leg 45 the box falls behind the elevation drive on up-legs, inside drive angle 90–150°, until it lags by 28.5°. The table's IMU elevation is unaffected; `motor_el_deg` is not a pointing there. Moved to pointing table v3, which fixes three defects this memo listed for v2.0: rows without a height era or phase (labelled by time interval, now by source file); `EL_SOLUTION_GLITCH` rows left `ok`, and the flag firing on passes through nadir; and the uncommitted generator with its pointer to a notebook in no commit.

Added the current-monitor comparison: the drive remains energised during the under-travel, but the six analysed frozen post-re-raise legs have no measurable running-scale draw. The background is approximately steady, downward-sweep draw declines gradually, and current alone does not exclude supply sag. Named the load thermistor and separated the stillness-detector boundary from the loss-of-response interval. All current numbers are regenerated by the supplied validator and checked against the reviewed reduction.

2026-10-23 Added cross-correlation phase analysis of the 07-17 raster sweeps (§7, Fig. 5). Measured the 3-D antenna phase center lever arm from the elevation axle ($z_{\rm pc} = +55.6 \pm 6.6$ cm along boresight, $x_{\rm pc} = +16.1$ cm along dipole arm, $y_{\rm pc} = -1.1$ cm transverse; total $\|\mathbf{r}_{\rm pc}\| = 57.9$ cm) and detected 11.3 cm of 2nd-harmonic gravitational flexure vs. elevation. Stationary platform wobble is 0.75 cm rms on < 5 s timescales. All numbers and figures regenerate via `validate_phase_center.py`.

A Reproducing this memo

Run `python make_products.py` in this directory, then `pdflatex memo-003.tex` twice. The script reads the inputs below, checks the SHA-256 of the one that is not tracked in git (the beam product's per-sample tooth amplitudes), and writes every figure in `figures/` and every table and quoted number in `tables/`. The tooth amplitudes are background-subtracted transmitter comb power from `empirical_raster_v0013`; they do not depend on pointing, and only their arm-0/arm-1 ratio is used. The beam fit's geometry corrections come from that product's `provenance.json`.

The current comparison is a *diagnostic* reduction of 72 correlator files. Run `python validate_current.py --raw-root /path/to/corr/files` before `make_products.py` to rebuild `validate_current.json`; an existing result is checked against its source and input hashes before its numbers become macros. The `inputs/current_selection.parquet` and `inputs/current_legs.csv` freeze the reviewed row/leg selections and refit IMU coordinate from notebooks 008 and 025 (canonical 025 commit `671451c`); they contain no current or temperature measurements. This IMU coordinate defines the current diagnostic's matched-angle window, not a replacement for the v3 pointing table. The validator reads all telemetry anew, uses `time_best` and disables index caching; it records the input-file hashes and checks every quoted current and temperature quantity against the canonical reduction.

The phase-center lever arm and flexure analysis (§7) is regenerated by `python validate_phase_center.py`, which reads the 07-17 raster files and writes `figures/phase_center.pdf` and `validate_phase_center.json`.

Running rows are interior rows of two-sided count changes with the other axis parked; parked rows have neither axis changing on either side. Each running row's background is the mean of nearby parked samples within five minutes, widened until at least two exist. Per-leg draw is the mean above that reference; drift and onset tests use whole legs, an early trend and its extrapolation uncertainty, with a lag-one correlation correction. The IMU-window contrast compares the late up-legs with the same window on earlier up-legs. Frozen-leg draws are pooled across the six selected legs; the comparison running leg is a continuous sweep that morning. Temperatures use the reviewed validity cuts and per-leg medians as onset markers. The monitored rail's exact wiring and load mix remain undocumented; no voltage, torque or motor-temperature inference is made. The current generator ran at memo source `4cbb38a`.

input	SHA-256 (first 16)
marjum-2026-07/curation/pointing_table.parquet	1e5fad8590f4ffeb
marjum-2026-07/imgs/fits/v0004_marjum_geometry/shared.json	8fcf5360cbfa14a0
marjum-2026-07/derived/beam/empirical_raster_v0013/dpss/diagnostics.npz	c829b29129c0ac6c
marjum-2026-07/derived/beam/empirical_raster_v0013/provenance.json	37b9f6b2432200c3
marjum-2026-07/derived/tx_feed_orientation/v0002/summary.json	1af458e39f88f320
memos/memo-003-marjum-2026-07-pointing/validate_current.json	5de383c7595442dd
memos/memo-003-marjum-2026-07-pointing/inputs/current_manifest.json	b36ebe06d5baca97
memos/memo-003-marjum-2026-07-pointing/inputs/current_selection.parquet	c86260c4e4ffebff
memos/memo-003-marjum-2026-07-pointing/inputs/current_legs.csv	b23fab0fa935c2bb
memos/memo-003-marjum-2026-07-pointing/validate_phase_center.json	2071ae9dc84018ba
pointing table version	v3+a29dd6f
eigsep (memo, campaign) commit	3a6ed5e
eigsep_base commit	
generated	2026-10-23 19:53 UTC